MIL-E-62140(AT)
27 August 1971
SUPERSEDING
DAPD-296B
24 February 1966

MILITARY SPECIFICATION

ENGINE, DIESEL: LIQUID COOLED, V-TYPE,
6-CYLINDER, 210 H.P.

1. SCOPE

1.1 Scope. This specification covers one type of 6-cylinder, V-type, liquid-cooled, 2-stroke-cycle, internal-combustion, compression ignition engine for use in military vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-A-548	-	Antifreeze, Ethylene Glycol, Inhibited.
O-I-490	-	Inhibitor, Corrosion, Liquid Cooling System.
P-C-437	-	Cleaning Compound, High Pressure (Steam) Cleaner.
VV-F-800	-	Fuel Oil, Diesel.

Military

MIL-P-514	-	Plates; Identification, Transportation-Data, and Blank.
MIL-L-2104	-	Lubricating Oil, Internal Combustion Engine (Heavy Duty).
MIL-L-10295	-	Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-C-11755	-	Compound, Antifreeze, Arctic-Type.
MIL-S-13488	-	Aircleaner, Engine: Heavy Duty, Oil-Bath Type (for Internal Combustion Engines).

FSC-2815

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MIL-S-13856 - Electrical Components for Automotive Vehicles; Waterproofness and Resistance to Temperature.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative.

STANDARDS
Federal
FED. TEST METHOD STD. NO. 791 - Lubricants, Liquid Fuels, and Related Products: Methods of Testing.

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-1400 - Engines, Gasoline or Diesel, Methods of Test.

DRAWINGS
Engineering Parts List
8738127 - Engine (Complete).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First production engine. The first engine produced from production tooling shall be furnished for inspection and tests. This engine shall be representative of the engines to be furnished under the contract.

3.1.1 Endurance. The engine using referee grade fuel shall operate for a period of 500 hours without requiring any more maintenance than changing oil, replacing oil filters, fuel filters and minor adjustments. At the conclusion of the endurance run, the engine shall develop not less than 90 percent of its rated horsepower.

3.1.1.1 Endurance run fuel. Engine shall be operated with diesel fuel conforming to the following: Using method No. 340-T of FED. TEST METHOD STD. NO. 791, referee grade diesel fuel shall have the following properties:

Flash point, minimum, °F.	100 or legal
Pour point, maximum, °F.	20

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Water and sediment, by volume, maximum, percent	.05
Distillation, 50% evaporated, minimum, °F.	500
90% evaporated, °F.	600 - 640
End point, °F.	650 - 690
Kinematic viscosity, centistokes, at 100°F.	1.6 to 4.5
Sulfur percent	.95 - 1.05
Corrosion	pass
Alkali and mineral acids	none
Cetane number	40 to 45

*The sulfur must be natural sulfur.

NOTE: The use of additives of any kind to produce a clean fuel, or solvent-treating the fuel, is not permitted.

3.2 Construction. Engine construction shall be in accordance with drawings listed or refered to in the Engineering Parts List 8738127.

3.2.1 Accessories and equipment. All electrical accessories and equipment, including wiring and electrical connections, shall conform to the applicable requirements of MIL-E-13856. Unless otherwise specified, all accessories and equipment shall be installed on the engine and properly adjusted.

3.3 Interchangeability of parts. With the exception of parts furnished in matched sets or for which a selective fit is specified, component assemblies, parts or service parts of the engine shall be so constructed to facilitate ease of installation, replacement and adjustment without requiring modification. Any design change that effects performance or changes the configuration of the above items requires prior USATACOM approval.

3.4 Steam and water jet cleaning. The engine and all its components shall withstand cleaning with high pressure steam and cleaner conforming to P-C-437 and water jet, without deterioration of seals or hose, water leakage past seals or gaskets, or other defects. Paint removal shall not be considered a deficiency. Steam and water jet cleaning is defined as follows:

> The jet is applied perpendicular to the surface being cleaned at a minimum distance of 1 ft. from the surface and cleaned at a rate of 1 square foot per minute (sq. ft./min). The minimum jet pressure shall be not less than 100 psi, nor more than 110 psi.

3.5 Production break-in run. Engine using fuel oil conforming to grade 2 of VV-F-800 shall be given a break-in run by the contractor to ensure the engine as submitted to the Government for acceptance shall meet all requirements specified herein.

3.5.1 Brake horsepower. At the completion of the production break-in run,

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the engine operating at 2800 rpm shall develop not less than 202 observed gross brake horsepower using specified fuel.

3.5.2 Operating conditions. Engine rating shall be corrected to the following conditions:

Dry air barometer	29.00 inches Hg.
Inlet air temperature	85°F.
Vapor pressure	0.38 in. Hg.
Dry air density	.0705 lbs/cu. ft.

3.5.3 Torque. At completion of the production break-in run and using specified fuel, the engine operating at 1600 rpm shall develop not less than observed torque of 420 pounds feet.

3.5.4 50-hour run. Fuel oil conforming to grade 2 of VV-F-800 shall be used during the 50-hour run.

3.6 Performance requirements.

3.6.1 Speed range. The engine shall operate satisfactorily under all loads and conditions as specified herein through a speed range of 1200 to 2800 revolutions per minute (rpm). The engine shall maintain a satisfactory idle speed of 650 to 700 rpm when using diesel fuel conforming to grade 2 of VV-F-800.

3.6.1.1 Governor. The governor shall limit the maximum engine speed as follows:

Full load operation	2785 to 2835 rpm
No load operation	2940 to 2990 rpm

The engine speed with no load on the engine shall stabilize within 3 seconds, after full rack position is reached.

3.6.1.2 Oil. Engine shall perform as specified herein using lubricating oil in accordance with the seasonal requirements of MIL-L-2104 from -10° to +115°F, and MIL-L-10295 from -65° to 0°F. The manufacturer may use oil of his own selection during engine build-up and the break-in run, or he may use oil conforming to grade 2 of MIL-L-21260.

3.6.2 Environmental requirements. The engine shall start within 2 minutes and shall demonstrate the performance characteristics specified herein under any of the following conditions or possible combination of conditions:

3.6.2.1 Temperature conditions.

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a. With integral cold starting aid, and without external aids after being cold-soaked, without benefit of solar radiation, to an ambient temperature of -25°F. Not less than 100 - 110 or max rpm cranking speed shall be demonstrated. (Cold-soak shall be defined as reducing the temperature of the engine its fuel, coolant and lubricating oil, to within 5°F of the ambient air temperature surrounding the engine.)
b. Without external aids, with cooling fan installed, and a fuel pressure of 5 psi at the inlet to the fuel injector supply pumps, when operated at the following temperatures with exposure to maximum solar radiation:

Ambient air temperature	115°F.
Intake air temperature (at air cleaner inlet)	115°F.
Coolant outlet temperature	230°F.
Oil sump temperature	At maximum temperature attainable up to 275°F.

3.6.2.2 Elevation conditions. At any elevation from sea level to 8,000 feet, except that the power requirements of the engine shall be reduced proportional to the elevation, so that at 8,000 feet elevation, the observed power output of the engine shall be not less than 75 percent of rated power.

3.6.2.3 Humidity conditions. Under relative humidity conditions as low as 5 percent at temperature of 115°F and as high as 100 percent at all temperatures from -25° to 85°F.

3.6.3 Coolant. The engine shall perform as specified herein using water and antifreeze compound conforming to O-A-548, arctic-type antifreeze compound conforming to MIL-C-11755, or water and corrosion inhibitor compound conforming to O-I-490.

3.6.4 Limiting operating temperatures. The engine shall not exceed the following limiting temperatures when operating at full load throughout the speed range of the engine under any of the conditions specified herein:

(1)	Oil sump	275°F.	(measured at the drain plug)
(2)	Coolant	230°F.	(measured at coolant outlet)

3.6.5 Grades and slopes. The engine shall start and demonstrate the performance characteristics specified herein when operated throughout the speed range of the engine, for not less than 30 minutes in each direction, with full and one quart low oil level conditions in the sump, on longitudinal grades or lateral slopes up to 60 percent in all directions, and as a result of said op-

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eration, no evidence of faulty lubrication, cooling, fuel supply, leakage, or other malfunction shall be found.

3.6.6 Exhaust smoke density. The maximum exhaust smoke density at full power position, when measured within three feet of the exhaust manifold outlet, shall not exceed the following conditions when using specified fuels.

Table I. Smoke density

Engine Speed RPM	Visual No.	Meter No.
1000	3	8
1400	3	7
2600	1	4

NOTE: The meter reading shall have precedence over the visual reading.

3.6.7 Fuel consumption. With engine operating at 2800 rpm and using fuel specified herein, the fuel consumption shall not exceed that specified in table II.

Table II. Maximum observed brake specific fuel consumption

RPM	Lbs Per BHP-HR
1800	.460
2800	.450

3.6.8 Oil consumption. After 25 hours of operation, the engine shall consume not more than 0.007 lbs. of lubricating oil per observed brake horsepower hour, when operating at full rack throughout the power check phase of the engine with oil sump temperatures of 180° to 260°F. The power check phase shall be determined by setting at full rack position and operating the engine at full load and engine speeds of 1600 - 2000 - 2400 - 2800 RPM's to assure meeting performance requirements.

3.6.9 Oil pressure. Under all operating conditions specified herein, including a one quart low oil sump condition, with an oil sump temperature of not less than 180°F, gallery oil pressure shall not be more than 80 pounds per square inch (psi) nor less than 32 psi, when the engine is operating at 2800 rpm, using specified oil and shall not be less than 5 psi when engine is idling between 650 and 700 rpm.

3.6.10 Electromagnetic compatibility. Unless otherwise specified, the engine shall meet the functional suppression requirements, class IIIA of MIL-STD-461, as specified.

3.7 Log sheet. The contractor shall maintain a log sheet of each engine run during all tests specified herein. The form of the log sheet and number

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of legible copies to be furnished to the Government shall be as approved by the procuring activity.

3.8 Exterior surface treatment. All exposed exterior surfaces of the engine and its components shall be cleaned, painted, or treated for corrosion resistance as specified on the applicable drawings, or if not so specified, in accordance with applicable provisions of MIL-STD-193.

3.9 Product identification.

3.9.1 Nameplate. Unless otherwise shown on the applicable drawings, the nameplate, data, and instruction plates shall conform to MIL-P-514.

3.10 Workmanship. Workmanship shall be of the quality necessary to produce engines free from defects that would affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. The contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer no later than 30 days prior to initiation of production for determination of acceptability. A written description of the system must be acceptable to the Contracting Officer prior to inspection for acceptance of any production engines. A maximum of 10 days is required by the contracting officer for determination of acceptability. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents reference herein.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan

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are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Inspection equipment. Inspection equipment drawings referenced by number herein are details of Inspection Equipment Drawing Lists-Inspection Equipment Drawings listed thereon will serve as standards for determining the adequacy of contractor's inspection equipment (see 6.10).

4.2 First production engine test. To determine conformance to 3.1 thru 3.1.1.1 after break-in run is completed, the first production engine shall meet all tests requirements specified herein. This test (see table III) and inspection shall be conducted by the Government at a place designated by the Government, and under Government supervision. During this test the data (see 6.8) shall be recorded after each hour of operation and at the completion of each period prior to stopping the engine (emergency stops excluded).

Table III. Endurance run

Period	Time, hours	R.P.M. or percent of max. rated speed	Percent of full rack load at indicated speed	Cylinder head temperature °F	** *Coolant Temp. °F
1	See 4.2.2				
2	2	500 rpm	0	Allow to vary with load	180
3	5	50 pct	75	"	180
4	1	500 rpm	0	"	180
5	5	75 pct	75	"	180
6	1	500 rpm	0	"	180
7	5	85 pct	75	"	180
8	1	500 rpm	0	"	180
9	5	100 pct	75	"	180
10	2	500 rpm	0	"	180
11	5	50 pct	75	"	180
12	1	500 rpm	0	"	180
13	5	75 pct	75	"	180
14	1	500 rpm	0	"	180
15	5	85 pct	75	"	180
16	1	500 rpm	0	"	180

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Table III. Endurance run (cont'd)

Period	Time, hours	R.P.M. or percent of max. rated speed	Percent of full rack load at indicated speed	Cylinder head temperature °F	** *Coolant Temp. °F
17	5	100 pct	75	Allow to vary with load	180
18	2	500 rpm	0	"	180
19	5	50 pct	75	"	180
20	1	500 rpm	0	"	180
21	5	75 pct	75	"	180
22	1	500 rpm	0	"	180
23	5	85 pct	75	"	180
24	1	500 rpm	0	"	180
25	5	100 pct	75	"	180
26	2	500 rpm	0	"	180
27	5	50 pct	75	"	180
28	1	500 rpm	0	"	180
29	5	75 pct	75	"	180
30	1	500 rpm	0	"	180
31	5	85 pct	75	"	180
32	1	500 rpm	0	"	180
33	5	100 pct	75	"	180
34	See 4.2.3.4 (powertest)				
35	2	500 rpm	0	"	180
36	5	50 pct	75	"	180
37	1	500 rpm	0	"	180
38	5	75 pct	75	"	180
39	1	500 rpm	0	"	180
40	5	85 pct	75	"	180
41	1	500 rpm	0	"	180
42	5	100 pct	75	"	180
43	2	500 rpm	0	"	180
44	5	50 pct	75	"	180
45	1	500 rpm	0	"	180
46	5	75 pct	75	"	180
47	1	500 rpm	0	"	180
48	5	85 pct	75	"	180
49	1	500 rpm	0	"	180
50	5	100 pct	75	"	180
51	2	500 rpm	0	"	180
52	5	50 pct	75	"	180
53	1	500 rpm	0	"	180
54	5	75 pct	75	"	180

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Table III. Endurance run (cont'd)

Period	Time, hours	R.P.M. or percent of max. rated speed	Percent of full rack load at indicated speed	Cylinder head temperature °F	**Coolant Temp. °F
55	1	500 rpm	0	Allow to vary with load	180
56	5	85 pct	75	"	180
57	1	500 rpm	0	"	180
58	5	100 pct	75	"	180
59	2	500 rpm	0	"	180
60	5	50 pct	75	"	180
61	1	500 rpm	0	"	180
62	5	75 pct	75	"	180
63	1	500 rpm	0	"	180
64	5	85 pct	75	"	180
65	1	500 rpm	0	"	180
66	5	100 pct	75	"	180
67	See 4.2.3.4 (powertest)				
68	2	50 pct	50	"	180
69	1	500 rpm	0	"	180
70	2	58.4 pct	100	"	180
71	2	58.4 pct	50	"	180
72	1	500 rpm	0	"	180
73	2	58.4 pct	100	"	180
74	2	58.4 pct	50	"	180
75	1	500 rpm	0	"	180
76	2	58.4 pct	100	"	180
77	2	58.4 pct	50	"	180
78	1	500 rpm	0	"	180
79	2	58.4 pct	100	"	180
80	2	58.4 pct	50	"	180
81	1	500 rpm	0	"	180
82	2	58.4 pct	100	"	180
83	2	58.4 pct	50	"	180
84	1	500 rpm	0	"	180
85	2	58.4 pct	100	"	180
86	2	58.4 pct	50	"	180
87	1	500 rpm	0	"	180
88	2	58.4 pct	100	"	180
89	2	58.4 pct	50	"	180
90	1	500 rpm	0	"	180
91	2	58.4 pct	100	"	180
92	2	58.4 pct	50	"	180

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Table III. Endurance run (cont'd)

Period	Time, hours	R.P.M. or percent of max. rated speed	Percent of full rack load at indicated speed	Cylinder head temperature °F	** *Coolant Temp. °F
93	1	500 rpm	0	Allow to vary with load	180
94	2	58.4 pct	100	"	180
95	2	58.4 pct	50	"	180
96	1	500 rpm	0	"	180
97	2	58.4 pct	100	"	180
98	2	58.4 pct	50	"	180
99	1	500 rpm	0	"	180
100	2	58.4 pct	100	"	180
101	2	58.4 pct	50	"	180
102	1	500 rpm	0	"	180
103	2	58.4 pct	100	"	180
104	2	75 pct	50	"	180
105	1	500 rpm	0	"	180
106	2	75 pct	100	"	180
107	4	75 pct	50	"	180
108	1	500 rpm	0	"	180
109	4	75 pct	50	"	180
110	1	500 rpm	0	"	180
111	4	75 pct	50	"	180
112	1	500 rpm	0	"	180
113	4	75 pct	100	"	180
114	1	500 rpm	0	"	180
115	4	75 pct	50	"	180
116	1	500 rpm	0	"	180
117	4	75 pct	100	"	180
118	1	500 rpm	0	"	180
119	4	75 pct	50	"	180
120	1	500 rpm	0	"	180
121	See 4.2.3.4 (powertest)				
122	4	75 pct	100	"	180
123	1	500 rpm	0	"	180
124	4	75 pct	50	"	180
125	1	500 rpm	0	"	180
126	4	85 pct	100	"	180
127	1	500 rpm	0	"	180
128	4	85 pct	75	"	180
129	1	500 rpm	0	"	180
130	4	85 pct	100	"	180

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Table III. Endurance run (cont'd)

Period	Time, hours	R.P.M. or percent of max. rated speed	Percent of full rack load at indicated speed	Cylinder head temperature °F	**Coolant Temp. °F
131	1	500 rpm	0	Allow to vary with load	180
132	4	85 pct	75	"	180
133	1	500 rpm	0	"	180
134	4	85 pct	100	"	180
135	1	500 rpm	0	"	180
136	4	85 pct	75	"	180
137	1	500 rpm	0	"	180
138	4	85 pct	100	"	180
139	1	500 rpm	0	"	180
140	4	85 pct	75	"	180
141	1	500 rpm	0	"	180
142	4	85 pct	100	"	180
143	1	500 rpm	0	"	180
144	4	500 rpm	75	"	180
145	1	500 rpm	0	"	180
146	4	100 pct	100	"	180
147	1	500 rpm	0	See note	
148	4	100 pct	100	"	180
149	1	500 rpm	0	"	180
150	4	100 pct	100	"	180
151	1	500 rpm	0	"	180
152	4	100 pct	75	"	180
153	1	500 rpm	0	"	180
154	4	100 pct	100	"	180
155	1	500 rpm	0	"	180
156	4	100 pct	75	"	180
157	1	500 rpm	0	"	180
158	4	100 pct	100	"	180
159	1	500 rpm	0	"	180
160	4	100 pct	75	"	180
161	1	500 rpm	0	"	180
162	See 4.2.3.4 (powertest)				
163	4	100 pct	100	"	180
164	1	500 rpm	0	"	180
165	4	100 pct	75	"	180

NOTE: Allow to vary with load, but not in excess of limits recommended by the engine manufacturer.

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Table III. Endurance run (cont'd)

Period	Time, hours	R.P.M. or percent of max. rated speed	Percent of full rack load at indicated speed	Cylinder head temperature °F	** *Coolant Temp. °F
166	1	500 rpm	0	See note	180
167	3	100 pct	75	"	180
168	1	500 rpm	0	"	180
169	3	100 pct	100	"	180
170	3	100 pct	75	"	180
171	1	500 rpm	0	"	180
172	3	105 pct	100	"	180
173	3	100 pct	75	"	180
174	1	500 rpm	0	"	180
175	3	100 pct	100	"	180
176	3	100 pct	75	"	180
177	1	500 rpm	0	"	180
178	3	100 pct	100	"	180
179	3	100 pct	75	"	180
180	1	500 rpm	0	"	180
181	2	100 pct	100	"	180
182	2	75 pct	75	"	180
183	1	500 rpm	0	"	180
184	2	85 pct	100	"	180
185	2	100 pct	100	"	180
186	1	500 rpm	0	"	180
187	2	75 pct	75	"	180
188	2.75	75 pct	100	"	180
189	2.75	85 pct	75	"	180
190	2.5	85 pct	100	"	180
191	2.5	85 pct	75	"	180
192	2.5	100 pct	100	"	180
193	2.5	100 pct	75	"	180
194	2.5	100 pct	100	"	180
195	2.5	100 pct	75	"	180
196	0.75	100 pct	100	"	180
197	0.25	500 rpm	0	"	180
198	0.75	100 pct	100	"	180
199	0.25	500 rpm	0	"	180
200	0.75	100 pct	100	"	180
201	0.25	500 rpm	0	"	180
202	0.75	100 pct	75	"	180

NOTE: Allow to vary with load, but not in excess of limits recommended by the engine manufacturer.

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Table III. Endurance run (cont'd)

Period	Time, hours	R.P.M. or percent of max. rated speed	Percent of full rack load at indicated speed	Cylinder head temperature °F	**Coolant Temp. °F
203	0.25	500 rpm	0	See note	180
204	0.75	500 rpm	100	"	180
205	0.24	500 rpm	0	"	210
206	0.75	100 pct	100	"	210
207	0.25	500 rpm	0	"	210
208	0.75	100 pct	100	"	210
209	0.25	500 rpm	0	"	210
210	0.75	100 pct	100	"	210
211	0.25	500 rpm	0	"	210
212	0.75	100 pct	75	"	210
213	0.25	500 rpm	0	"	210
214	0.25	100 pct	75	"	210
215	0.25	500 rpm	0	"	210
216	0.75	100 pct	100	"	210
217	0.25	500 rpm	0	"	210
218	0.75	100 pct	100	"	210
219	0.25	500 rpm	0	"	210
220	0.75	100 pct	100	"	210
221	0.25	500 rpm	0	"	210
222	0.75	100 pct	100	"	210
223	0.25	500 rpm	0	"	210
224	0.75	100 pct	100	"	210
225	0.25	500 rpm	0	"	210
226	0.75	100 pct	100	"	210
227	0.25	500 rpm	0	"	210
228	0.75	100 pct	100	"	210
229	0.25	500 rpm	0	"	210
230	0.75	100 pct	100	"	210
231	0.25	500 rpm	0	"	210
232	0.75	100 pct	100	"	210
233	0.25	500 rpm	0	"	210
234	0.75	100 pct	100	"	210
235	0.25	500 rpm	0	"	210
236	0.75	100 pct	100	"	210
237	0.25	500 rpm	0	"	210
238	0.75	100 pct	100	"	210

NOTE: Allow to vary with load, but not in excess of limits recommended by the engine manufacturer.

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Table III. Endurance run (cont'd)

Period	Time, hours	R.P.M. or percent of max. rated speed	Percent of full rack load at indicated speed	Cylinder head temperature °F	** *Coolant Temp. °F
239	0.25	500 rpm	0	See note	210
240	0.75	100 pct	100	"	210
241	0.25	500 rpm	0	"	210
242	0.75	100 pct	100	"	210
243	0.25	500 rpm	0	"	210
244	0.75	100 pct	100	"	210
245	0.25	500 rpm	0	"	210
246	.50	110 pct	0	"	210
247	See 4.2.3.4 (powertest)				

NOTE: Allow to vary with load, but not in excess of limits recommended by the engine manufacturer.

4.2.1 Failure. Failure of a first production engine to conform to any of the requirements specified, or any major deficiency of a workmanship or material nature found as a result of the test shall be cause for rejection, and refusal to continue acceptance of production engines until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Correction of any deficiency must be accomplished by the manufacturer at no cost to the Government on all engines accepted under the contract. Any deficiencies found as a result of test is considered prima facie evidence that engines already accepted are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that no deficiencies exist.

4.2.2 Endurance test. The engine shall be operated in accordance with table III to determine conformance to 3.1.1.1.

4.2.3 Acceptable endurance performance. During the endurance test, each engine shall be observed for compliance with the following performance requirements during the power check tests (see 4.2.3.4).

Speed range	(see 3.6.1)
Governed speeds	(see 3.6.1.1)
Brake horsepower	(see 3.5.1)
Torque	(see 3.5.3)
Exhaust smoke density	(see 3.6.7)
Fuel consumption	(see 3.6.8)
Oil consumption	(see 3.6.9)
Oil pressure	(see 3.6.10)
Engine life	(see 3.1.1)

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4.2.3.1 Cold starting. To determine conformance to item "a" of 3.6.2.1, the engine, with integral winterization aids, shall be cold started at -25°F. on fuel specified in 3.1.1.1. Prior to the cold start, the engine shall be cold-soaked in an ambient temperature of -25°F. for a period of 24 hours. After each start, the engine must operate for a sufficient time to stabilize engine operation.

4.2.3.2 High temperature starting. The engine, without aids, shall be placed in a high temperature chamber maintained at a temperature of 115°F., and operated at full load until all temperatures are stabilized within the limits specified in 3.6.4. The engine shall then be shut down; hot soaked, and restarted when the fuel temperature at the inlet to the injection pump reaches its maximum value. Fuel shall be supplied to the engine mounted fuel supply pump at a pressure of 5 psi. The fuel pressure shall be shut off during the hot soak period. The starting time shall not exceed 2 minutes, to determine conformance to item "b" of 3.6.2.1.

4.2.3.3 Slope test. The engine shall be tested to determine conformance to 3.6.5.

4.2.3.4 Power check test. Full rack performance characteristics of the engine shall be determined by operating the engine according to the following schedule: (See table III)

Period	Engine Speed	Rack Position	Fuel
1	1600	Full Rack	(See 3.1.1.1)
2	2000	"	"
3	2400	"	"
4	2800	"	"

All data necessary to determine the performance characteristics specified in 4.2.3 or 4.6.3, as applicable, shall be recorded in the log. Performance curves showing horsepower, torque and specific fuel and oil consumption shall be required in addition to the log sheets. Oil consumption data shall be obtained from endurance hours. Operating conditions 3.5.2 shall be maintained during test, as specified in MIL-STD-1400 (see 6.3).

4.2.3.4.1 Servicing. Prior to the power check test, the lubricating oil shall be changed, all filters serviced, and valve clearances checked and adjusted to applicable requirements.

4.2.3.4.2 Atmospheric conditions. The barometric pressure shall be recorded during the power check test.

4.2.3.5 Elevation test. The engine shall be placed in a simulated environ-

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mental condition of 8,000 feet elevation, and its general performance characteristics observed to determine conformance to 3.6.2.2.

Elevation feet	Pressure inches of* mercury	Temperature °F.
8,000	22.2	90

*NOTE: Measured at air cleaner inlet and exhaust outlet.

4.2.3.6 High temperature operational test. To determine compliance with item of 3.6.2.1, the last 100 hours of the 500 hour endurance test shall be conducted with the intake air temperature of 115° ± 5°F at an ambient air temperature of 115° ± 5°F and a fuel temperature of 115° ± 5°F at the fuel transfer pump inlet. The power check tests shall be made at standard operating conditions.

4.2.3.7 Steam and water jet cleaning test. At the completion of the 500 hour endurance test, and just prior to the tear down inspection, the engine shall be subjected to normal steam cleaning with cleaner as specified in P-C-437. Immediately upon completion of the steam cleaning operation, the engine shall be subjected to normal water jet cleaning as specified herein. Prior to cleaning, the air intake opening, flywheel, exhaust, and all other openings that might be affected shall be covered. The engine shall start and operate at idle speed without misfiring within two minutes immediately after the completion of both the steam and water jet operations. During the disassembly of the engine, special notice should be made of water entering the engine through joints, seals, gaskets, etc. Evidence of malfunction or of water seepage past seals, gaskets, joints, etc., shall constitute a failure of the test.

4.2.3.8 Electromagnetic compatibility test. To determine conformance to 3.6.11, engine shall be tested for meeting electromagnetic compatibility requirements specified (see 6.10).

4.3 Sampling for lot acceptance and testing.

4.3.1 Inspection lot. A lot shall consist of all engines of one type from an identifiable production period, submitted at one time for acceptance. Each sample engine shall be selected from the lot in accordance with inspection Level II of MIL-STD-105 and shall be tested in accordance with 4.5.

4.3.2 Examination. Each of the engines selected shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table IV.

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Table IV. Classification of defects

	Characteristics	Major AQL's 6.5 Defects Period Units	Gage or other method of inspection
101.	Valve tappet clearance, each valve	Exhaust valves improperly adjusted	Gage
102.	Oil sump, fuel and oil filters	Dirt, chips and foreign particles, due to production	Visual & functional
103.	Fuel, oil, and coolant leaks	Leakage	Visual
104.	Fuel lines	Damaged	Visual & functional
105.	Governor	Malfunction	Visual & functional
106.	Torque on cylinder head bolts, intake and exhaust manifold flange bolts, vibration damper	Improper torque	Torque wench
107.	Crankshaft	Excessive or restrictive end play	Gage & functional
108.	Engine	Malfunction	Functional
109.	Fuel system components	Malfunction - damage or leaks, etc.	Visual & functional
110.	Fuel injection system, timing and components	Malfunction, improper adjustment	Gage
111.	Air cleaner	Malfunction	Functional
112.	Air box pressure	Exceeds limits	Gage
113.	Crankcase pressure	Exceeds limits	Gage
		Minor AQL 25 Defects per 100 units	
201.	Linkage	Improperly adjusted	Visual
202.	Minor assemblies	Omitted	Visual
203.	Engine	Improper adjustment or installation of components	Visual & functional
204.	Fuel lines, hose, vents, shut-off valve	Improper assembly or installation	Visual
205.	Painting	Improper application	Visual
206.	Workmanship	Not following good practice, improper installation or adjustment of components	Visual & functional

4.3.2.1 Unclassified characteristics. All defects having no bearing on function, safety, interchangeability or life, but which are considered depar-

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ture from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots or 10 lots within a 30 day period will be added to the minor classification of defects of the specification with no increase in AQL's.

4.4 Production break-in run. Each engine shall be operated as specified herein. Any unsatisfactory operation or conditions which require correction shall be corrected by the contractor before the engine is presented for acceptance.

4.4.1 Tear down inspection. At the conclusion of the break-in run, the contractor in the presence of the Government inspector shall disassemble and inspect the first 10 engines, and every second engine of the next 10 of any contract except overlapping contracts. Disassembly shall be accomplished to the extent required to perform the following inspection. Any engine not requiring replacement parts shall be reassembled and subjected to the acceptance power check test only.

a. Oil contamination
b. Dirt, chips or foreign matter in the engine block, oil pan, filters, and accessory drive case
c. Main bearing bolt torque
d. Connecting rod bolt torque
e. Cylinder bore scuffing, scoring, galling, etc.
f. Piston scuffing and burning

4.4.1.1 Corrections and reassembly. Should there be evidence of any condition of engine or parts requiring correction, such corrections shall be made by the contractor prior to reassembly of the engine and resubmission to the break-in run. In addition, the contractor shall submit objective evidence to the Government, that corrective action has been taken in production to eliminate any deficiency disclosed. Failure of the contractor to submit such evidence shall be cause for refusal by the Government to continue acceptance of subsequent engines. Any engine requiring replacement parts shall be reassembled and subjected to the break-in run and acceptance power check test.

4.5 Acceptance tests and inspection. Each engine submitted to the Government shall be subjected to the applicable tests specified in table V, and shall be observed for compliance with the performance data herein. Engines submitted to the Government for acceptance shall be observed for compliance with the following performance requirements upon completion of the break-in run.

Speed range	(see 3.6.1)
Governor setting	(see 3.6.1.1)
Brake horsepower	(see 3.5.1)
Torque	(see 3.5.3)
Oil pressure	(see 3.6.10)

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4.6 Fifty-hour control test. A 50-hour quality control test specified in 3.5.4 and table VI, shall be conducted by the contractor as specified herein for the purpose of determining the continued quality of the engine. Each engine selected in accordance with 4.6.1 shall have completed the examination and the acceptance test specified herein. Inlet air temperatures shall not be less than 70° nor more than 100°F. The lubricating oil temperature shall not exceed 260°F. The coolant temperature shall be maintained at 180° ± 5°F.

Table V. Location for tests

Title	Paragraph No.	Place of Manufacture	Government Laboratory
First production engine test	4.2		X
Endurance test	4.2.2		X
Cold starting	4.2.3.1		X
High temperature starting	4.2.3.2		X
Slope test	4.2.3.3		X
Power check test	4.2.3.4		X
Servicing	4.2.3.4.1		X
Elevation test	4.2.3.5		X
High temperature operational test	4.2.3.6		X
Steam and water jet cleaning test	4.2.3.7		X
Radio noise suppression test	4.2.3.8		X
Examination	4.3.2	X	
Production break-in run	4.4	X	
Tear down inspection	4.4.1	X	
Corrections and reassembly	4.4.1.1	X	
Acceptance tests	4.5	X	
50 hour control test	4.6	X	
Selection of test sample	4.6.1	X	
Control test failure	4.6.2	X	
Performance	4.6.3	X	
Corrections	4.6.4	X	
Disposition of engine	4.6.5	X	

Table VI. 50 hour control test

Period	Time (hrs)	Speed (rpm)	Rack Position (% of full rack)	Fuel
1	(run in schedule) - (See table I)			
2	5	1600	50	Fuel conforming to grade 2 of VV-F-800 shall be used during the entire run.
3	5	2000	75	
4	5	2800	75	
5	5	2400	100	
6	30	2800	100	
7	Power check test (see 4.2.3.4)			

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Coolant Temperature: The coolant temperature shall be maintained at 180° ± 5°F.

NOTE: During this test, the data (see 6.9)shall be recorded after each hour of operation at the completion of each period prior to stopping the engine.

4.6.1 Selection of test sample. Engines shall be selected at the rate of one per month when production is 100 units per month or less, and at the rate of two per month when production is greater than 100 units per month. Engines shall be identified as to production period.

4.6.2 Control test failure. Failure of a control test sample to pass any specified examination or test shall be cause for the Government to refuse acceptance of subsequent lots until it has been proven to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.6.3 Performance test. The engine shall be observed for conformance to the following performance requirements (see 3.6) at the completion of the test specified in 4.6 and shall be reinspected for the deficiencies classified in 4.3.2.

Speed range	(see 3.6.1)
Governor setting	(see 3.6.1.1)
Brake horsepower	(see 3.5.1)
Torque	(see 3.5.3)
Exhaust smoke density	(see 3.6.6 and 6.6)
Fuel consumption	(see 3.6.7)
Oil consumption	(see 3.6.8)
Oil pressure	(see 3.6.9)
Limiting operating temperatures	(see 3.6.4)

4.6.4 Corrections. Upon completion of the test specified in 4.6, the engine shall be partially disassembled as in 4.4.1. Evidence of component or parts damage indicated by this inspection shall require such further disassembly of the engine necessary to determine the cause of failure and correction.

4.6.5 Disposition of engine. After correction of all defects disclosed by the test, engine shall be reassembled, tested as specified in 4.5, inspected, and if found acceptable, shipped as a regular production engine.

4.7 Inspection for preparation of delivery. Inspection of preservation, packaging, packing, and marking shall be in accordance with the applicable procedures outlined in section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable packaging data sheet as furnished at the time of invitation for bids; for the level of preservation and packaging, and level of packing specified (see 6.2).

6. NOTES

6.1 Intended use. The engines covered by this specification are intended for production use, as spares, or as replacements for military tactical and transport vehicles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Engineering Parts List or Drawing number (see 3.2).
(c) Address to which first production sample shall be sent (see 4.2).
(d) Form and number of copies of log sheet to be furnished for production runs and acceptance tests (see 3.7).
(e) Additional data for nameplate, when required (see 3.9.1).
(f) Selection of method and applicable level of preservation and packaging, and level of packing of the referenced specification (see 5.1).

6.3 Horsepower. Observed brake horsepower shall be corrected to standard atmospheric conditions according to the following:

$$\text{Corrected bhp} = \text{observed bhp} \times \frac{29.00}{B - e} \times \frac{460 + t}{545}$$

where

B = test barometric pressure, in. Hg wet
e = water vapor pressure, in. Hg
t = intake air temperature, °F. (see 5.2.2 of MIL-STD-1400)

6.4 Referee grade oil. The following designations apply for the various grades (see 3.6.1.2):

Grade	Government Designation	Mfg. Designation
10	MB2	CHL-140
30	MB1	CHL-139
50	None	None

6.5 Brake horsepower. Brake horsepower shall be defined as the horsepower output of the engine at the flywheel.

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6.6 Exhaust smoke density. Exhaust smoke density shall be determined with a Robert Bosch Model EFAW 66 Smoke Meter or equal (as approved by responsible engineering activity) and Model EFAW 65 Sampling Pump or equal (as approved by responsible engineering activity). The following table may be used to visually define the degree of exhaust smoke density:

Description of Exhaust Smoke	Classification
Clear	1
Haze	2
Light Gray	3
Medium Gray	4
Dark Gray to Black	5

NOTE: Observation of exhaust smoke should be made against a white background.

6.7 Endurance run. The following data shall be recorded at the end of each hour of operation, at the completion of each period, and just prior to stopping the engine (emergency stops excluded) (see 4.2).

a. Engine speed, rpm.
b. Observed brake horsepower (bhp).
c. Observed torque (lbs-ft).
d. Air box pressure (In. Hg. gage).
e. Exhaust manifold pressure, in. of water.
f. Fuel pressure after secondary filter, psi.
g. Fuel supply pressure at transfer pump, psi.
h. Lubrication oil pressure, gallery, psi.
i. Crankcase pressure, inches of water.
j. Lubricating oil temperature, ^{o}F (sump).
k. Lubricating oil temperature inlet, ^{o}F, gallery.
l. Fuel temperature after secondary filter, ^{o}F.
m. Air temperature at air cleaner inlet, ^{o}F.
n. Air temperature air box ^{o}F.
o. Coolant temperature, inlet and outlet.
p. Exhaust temperature ^{o}F.
q. Test cell ambient air temperature.
r. Blowby, cfm.
s. Fuel flow, lbs/hr.
t. Specific fuel consumption, lbs/bhp-hr.
u. Specific oil consumption, lbs/bhp-hr (at power check).
v. Barometric pressure of test cell - once each 4-hour period.
w. Exhaust smoke density (see 6.6).

6.8 50 hour control data. All data shall be obtained under stabilized

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operating conditions, with engine coupled to a dynamometer. At each start, the engine shall be warmed up until torque, speed, and temperatures have been stabilized for a period of not less than 1 minute. All applicable temperatures shall be recorded at the conclusion of each warm-up period. Warm-up starting, and stopping times shall be recorded; however, warm-up time and running time of less than 30 minutes, unless specified in table IV, shall not be counted toward fulfillment of test run hours. Prior to test, all applicable settings and adjustment, such as valve lash, etc., shall be inspected and data recorded. All adjustments and setting shall be reset to specifications before test is begun. The following data shall be recorded (see 4.6):

a. Engine speed, rpm.
b. Observed brake horsepower, bhp.
c. Observed torque lb-ft.
d. Air box pressure (in. Hg. gage).
e. Fuel pressure after secondary filter, psi.
f. Fuel supply pressure at inlet to engine driven supply pump, psi.
g. Lubricating oil pressure, psi.
h. Lubricating oil temperature in oil sump, ^{o}F.
i. Fuel temperature after secondary filter, ^{o}F.
j. Coolant temperature - Engine inlet and outlet, ^{o}F.
k. Air temperature at air inlet, ^{o}F.
l. Cell ambient air temperature, ^{o}F.
m. Exhaust temperature.
n. Fuel flow, lb/hr.
o. Brake specific fuel consumption, lb/obs. bhp-hr.
p. Specific oil consumption, lb/obs. bhp-hr.
q. Exhaust smoke density.

6.9 Measuring and inspection of testing equipment. When inspecting the suppliers inspection equipment, the Government inspector will determine that the supplier has available and utilizes correctly, gaging, measuring, and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The supplier will have available a set of master gages, standards, and appropriate instruments for calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the supplier and made available for review by the Government. Periodically, calibration of gages, standards, and inspection equipment will be checked by authorized Government personnel to determine compliance with gage requirements (see 4.1.3).

6.10 Electromagnetic compatibility test. The contracting officer should submit a copy of the report(s) required in 4.2.3.8 to the Commanding General,

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U. S. Army Electronics Command, ATTN: ANSEL-RD-GF, Ford Monmouth, New Jersey for approval.

Custodian:
Army - AT

Preparing activity:
Army - AT
Project No. 2815-A065

User:
Army - EL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-E-62140(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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